EFTHIMIOS VLAHOS

Senior Machine Learning Engineer | Applied AI / Forward-Deployed Engineer | AI Architect | LLM Agents, RAG, Evals
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SUMMARY

Senior Machine Learning Engineer with 6+ years building production AI systems and the last three shipping **LLM agents, document AI and evaluation pipelines** in regulated industries (insurance, energy, manufacturing). I take AI systems from discovery to production and own them there: stakeholder scoping, architecture, **self-hosted and API LLMs**, hybrid LLM + rules decision engines, evals and guardrails, deployment and reliability. Co-founder of an AI platform company with paying customers. The part of the work I enjoy most is between the demo and production: sitting with the people who own the problem until it is precisely defined, then building the system and staying accountable for it once it ships. Targeting Forward-Deployed / Applied AI Engineer, AI Architect and AI engineering leadership roles.

EXPERIENCE

Senior Machine Learning Engineer — Alliant Insurance Services, Manhattan, NY (Hybrid) Feb 2026 – Present

- Delivered **40% straight-through processing** on ~65K insurance policy submissions in 7 weeks (97% AI coverage, **0.10% true reviewer reversals**, **$0.0085 per policy**, ~870 reviewer hours saved) by building a **hybrid LLM + deterministic validation engine** integrated with Salesforce and replaying every new rule against previously rejected cases before release.
- Raised document extraction accuracy from **91.5% to 93.9%** and cut per-document LLM cost to **$0** with data kept on-prem by migrating production from Azure GPT-5.1 to a **self-hosted Qwen3.5 MoE vision model on vLLM** (NVIDIA A100s): built the 39-document ground-truth set and field-level comparator, ran a six-model bake-off and kept a one-command rollback.
- Designed and shipped a **commercial-submission engine in two weeks** as the in-house alternative to a $120K vendor (**12 releases in 9 days**, 99.3% field accuracy with **0 fabricated figures**, 1,700+ tests, zero-failure load test) by building deterministic email intake, document classification, **citation-backed extraction**, a rules-as-data checklist, a number-guarded LLM narrative, React 19 UI and machine API.
- Cut reviewer error on scanned-mail cancellation notices from **10.1% to 2.7%** (missed cancellations 0.22%) while turning **26K+ documents** into ~4,500 confirmed cancellations at $0 model cost, by building a **text-layer-first document classifier** (94% of documents decided in under 0.1 s), vision-LLM extraction for the rest and a **replay simulator** that proves every fix with zero regressions and zero model calls.
- Turned a **2-day silent outage class into ~2-minute auto-recovery** (and a nightly 36-minute event-loop freeze into 4–6 seconds) on a shared **8x A100 on-prem AI platform** by designing a **watchdog sidecar** (real-inference probes, busy-vs-dead detection, restart-storm breaker), a scripted deploy tool with in-image tests and **one-command rollback**, and advisory-lock async fixes.
- Proved that **relational (graph) risk features** lift an underwriting **risk-score XGBoost model** (significant out-of-time AUROC gain, P=0.993, **3x larger on new business**; next-year claim rate **10.6% to 30.4%** across property-manager-book deciles) by building a **250K account-year leakage-safe panel**, splink entity resolution (23K vendor codes to 16K entities), bootstrap confidence intervals and SHAP attribution; recommended three explainable monotone features over a GNN for **NAIC / NY-DFS** governance.
- Ran stakeholder discovery and executive reporting as the team's hands-on partner to product owners and the CIO: analyzed **100K+ customer-service cases** with on-prem LLM intent classification (**49/50 hand-audit agreement**, queue-guarded so production kept priority), showed that 52–61% of resident emails were policies to upload and set the build order; delivered CIO accuracy reports reconciled to live Salesforce data **22/22 days**.

Machine Learning Engineer — Con Edison, Manhattan, NY (Hybrid) Jan 2025 – Jan 2026

- Cut ad-hoc reporting time **40%** for **100+ analysts** and non-technical stakeholders by architecting a **LangGraph text-to-SQL agent** that routes natural-language requests with semantic schema selection (query embeddings pick the tables), generates and runs the SQL and produces the report.
- Drove **$2M+** in Energy Efficiency campaign revenue at **85%** precision by engineering the **propensity-to-enroll pipeline on Databricks**: Delta Live Tables ingestion, PySpark and Feature Store features, MLflow experiments with a champion **XGBoost** model, Unity Catalog registry and batch inference on Databricks Jobs.
- Lifted entity-match F1 **35%** over rules-based matching across **20+ source systems** by building a hybrid **entity-resolution pipeline**: fine-tuned Sentence Transformers embeddings, FAISS approximate-nearest-neighbor clustering and **LLM few-shot classification as the contextual judge** for final matching.
- Reduced model onboarding time **50%** by building the **MLOps platform on Azure** (MLflow, Docker, Terraform, Kubernetes with Kubeflow Pipelines and KServe, Jenkins, Prometheus/Grafana) that standardized training, registry, deployment and observability.

Data Scientist (Contract) — Cornerstone Building Brands, East Meadow, NY (Remote) May 2024 – Mar 2025

- Generated **$3.5M** incremental revenue by designing and productionizing a **recommendation system on Databricks** across **10,000+ SKUs** for dealer and contractor customers (scikit-learn, feature store, MLflow tracking and registry).
- Increased quote-to-close conversion **20%** by deploying a production **Flask prediction service** (Docker, Kubernetes) that serves tuned logistic-regression and random-forest models in real time.

- Cut pipeline runtime **35%** and excess inventory costs **15%** by building **ETL/ELT pipelines** in Databricks (PySpark, Delta Lake) from Azure Data Lake with Airflow and GitHub Actions CI/CD; Power BI dashboards for **50+** executives and plant managers.

Data Analyst (Contract) — Microsoft, East Meadow, NY (Remote) Oct 2020 – May 2024

- Cut LLM chatbot hallucination rates **30%** and raised user satisfaction **25%** by analyzing enterprise-scale datasets (Python, SQL, R) with rigorous anomaly detection and data cleansing.
- Cut manual escalations **20%** and improved contextual accuracy for key clients by fine-tuning a **Phi small language model** (PyTorch, TensorFlow) and building RAG chatbot prototypes A/B-tested on **5,000+ conversations**.

ENTREPRENEURIAL EXPERIENCE

Co-Founder & AI Platform Lead — Pyxeon (side venture), Long Island, NY · pyxeon.com Mar 2026 – Present

- Ship customer-requested changes with a median approved-plan-to-deploy time **under 24 hours** at **under $10 agent cost per change** by building an **autonomous coding pipeline**: Claude Sonnet 5 plans each request from a tier-prioritized queue; on approval an agent implements, tests and opens a PR; an **independent Opus 5 verifier** reviews the diff against the plan; every change ships with an evidence bundle and **one-click rollback**.
- Run **8 live services** at **99.9% monitored uptime** by architecting the platform with **multi-tenant isolation** (per-customer Supabase databases, encrypted GitHub tokens), Clerk auth and Stripe subscription billing with per-tier quotas.
- Scale agent runs and self-heal outages by designing the **Docker agent-scaling architecture** (Redis job queue, dispatcher launching resource-limited containers with HMAC-signed callbacks and a watchdog, Kubernetes job manifests) and an operations loop in which uptime checks open a fix request after 15 minutes of downtime and route it through the same pipeline.
- Won **two paying customers** with four customer-facing apps in production by owning product, pricing and go-to-market: tiered subscriptions with request quotas, an **AI consultation funnel** and an **App Factory** that turns a business intake into a branded application; **1,500+ backend and 47 frontend tests**, ~600 commits since March 2026.

SELECTED PROJECTS

- **Trader Hub** — AI options and equities trading platform on Alpaca: **LangGraph strategy council** of six specialist agents, **graduation gate** (Sharpe at least 1, drawdown at most 15%) before live trading, lookahead-safe backtests; **323 tests**.
- **Blueprint Takeoff** — **vision-LLM application** that reads an electrical drawing's legend, counts every symbol across multi-page PDFs and produces a material takeoff in under a minute; Docker.
- **DeskSergeant** (independent consulting) — **multi-tenant fraud and AI-security platform** (FastAPI, Next.js 16, Neon Postgres, iOS); **1,287 backend tests**; 182 architecture decision records.

TECHNICAL SKILLS

Agentic AI: multi-agent orchestration (planner / implementer / verifier), LangGraph (checkpointing, human-in-the-loop interrupts), Claude Agent SDK, OpenAI Agents SDK, function calling, structured outputs, MCP (Model Context Protocol), A2A, context engineering, prompt caching, sandboxed code execution, browser and computer-use agents, durable workflows

Agent Evaluation & Safety: ground-truth fixtures, trajectory and tool-call evals, LLM-as-judge with judge calibration, simulate-before-ship rule replays, regression suites in CI, guardrails (number guards against fabricated figures, prompt-injection defense), cost ceilings, model routing

Retrieval & Knowledge: RAG and agentic RAG (chunking, hybrid BM25 + dense retrieval, cross-encoder reranking, grounding and citations), GraphRAG, Sentence Transformers, FAISS, Chroma, pgvector, Qdrant, entity resolution (splink), text-to-SQL

Models, Fine-Tuning & Serving: Claude (Anthropic API), Qwen (MoE, vision), Phi, Llama; post-training: SFT, LoRA/QLoRA, DPO/ORPO, GRPO/RLVR, quantization; Hugging Face TRL, Unsloth, Axolotl; **vLLM on NVIDIA A100s** (tensor parallelism, prefix caching), LiteLLM; PyTorch, TensorFlow, scikit-learn, XGBoost, SHAP

MLOps & AgentOps: MLflow, Databricks (Delta Live Tables, Feature Store, Unity Catalog, Jobs), GitHub Actions, Prometheus/Grafana, Datadog, LLM tracing (LangSmith, Langfuse, Arize Phoenix, OpenTelemetry), self-healing watchdogs, scripted deploys with one-command rollback

Cloud, Data & Enterprise Integration: Azure (Machine Learning, OpenAI, Data Lake, AKS), AWS (SageMaker, S3, EKS), on-prem NVIDIA DGX, Spark/PySpark, Delta Lake, Airflow, PostgreSQL, SQL Server, Redis, Salesforce (CometD platform events, SOQL, Reports API)

Engineering: Python (asyncio), TypeScript, C#/.NET 8, SQL, R; FastAPI, Flask, React 19/Next.js, SQLAlchemy, JWT/Okta/Clerk auth, Stripe; pytest, Vitest, Playwright, test-first development, architecture decision records, CI/CD, AI-assisted engineering (Claude Code)

EDUCATION

Stony Brook University — Master of Science, Applied Mathematics & Statistics May 2023

CUNY Hunter College — Bachelor of Science, Mathematics & Physics; Minor in Computer Science May 2020

CERTIFICATIONS

Databricks Certified Data Engineer Associate (May 2025) | **AWS Certified Cloud Practitioner** (Apr 2024) | **LLMOps** and **MLOps**, Duke University (2024)